

**Lecture-31**  
**Mathematics 1 (15B11MA111)**

**CO [C105.5]**

**Module: Laplace Transforms**

# Topics to be covered

- ❖ **Solution of initial value problems.**

## **Reference for the lecture**

**R.K Jain and S.R.K. Iyenger**, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

## Example

Solve the following initial value problem using Laplace Transform

$$y'' - 4y' + 3y = 6t - 8, \quad y(0) = 0, \quad y'(0) = 0$$

**Solution:** Taking the Laplace transform of DE

$$L\{y'' - 4y' + 3y\} = L\{6t - 8\}$$

The Laplace transform is linear and we have

$$L\{f^{(n)}(t)\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 0, 1, 2, 3, \dots$$

$$\therefore L\{y''\} - 4L\{y'\} + 3L\{y\} = 6L\{t\} - 8L\{1\}$$

$$s^2 L\{y\} - sy(0) - y'(0) - 4[sL\{y\} - y(0)] + 3L\{y\} = 6\frac{1}{s^2} - 8\frac{1}{s}$$

$$\text{or } s^2 L\{y\} - 4sL\{y\} + 3L\{y\} = 6\frac{1}{s^2} - 8\frac{1}{s}$$

$$(s^2 - 4s + 3)L\{y\} = 6\frac{1}{s^2} - 8\frac{1}{s}$$

$$L\{y\} = 6\frac{1}{s^2(s^2 - 4s + 3)} - 8\frac{1}{s(s^2 - 4s + 3)}$$

$$y = L^{-1} \left\{ 6\frac{1}{s^2(s^2 - 4s + 3)} - 8\frac{1}{s(s^2 - 4s + 3)} \right\}$$

$$y = 6L^{-1} \left\{ \frac{1}{s^2(s^2 - 4s + 3)} \right\} - 8L^{-1} \left\{ \frac{1}{s(s^2 - 4s + 3)} \right\} \quad (1)$$

$$\frac{1}{s^2(s^2 - 4s + 3)} = \frac{1}{s^2(s-3)(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-3} + \frac{D}{s-1}$$

As (s-3) and (s-1) are linear factors, we have

$$C = \frac{1}{s^2(s-1)} \Big|_{s=3} = \frac{1}{3^2(3-1)} = \frac{1}{18}$$

$$D = \frac{1}{s^2(s-3)} \Big|_{s=1} = \frac{1}{1^2(1-3)} \Big|_{s=1} = -\frac{1}{2}$$

To determine  $A$  and  $B$ , we have

$$1 = As(s^2 - 4s + 3) + B(s^2 - 4s + 3) + Cs^2(s-1) + Ds^2(s-3)$$

$$1 = As(s^2 - 4s + 3) + B(s^2 - 4s + 3) + \frac{1}{18}s^2(s-1) - \frac{1}{2}s^2(s-3)$$

Comparing the coefficients of  $s^3$  and constant term both sides, we get

$$0 = A + \frac{1}{18} - \frac{1}{2}, \quad 1 = 3B$$

$$\therefore A = \frac{1}{2} - \frac{1}{18} = \frac{9-1}{18} = \frac{4}{9}, \quad B = \frac{1}{3}$$

$$\frac{1}{s^2(s^2 - 4s + 3)} = \frac{4}{9} \frac{1}{s} + \frac{1}{3} \frac{1}{s^2} + \frac{1}{18} \frac{1}{s-3} - \frac{1}{2} \frac{1}{s-1}$$

$$L^{-1}\left\{\frac{1}{s^2(s^2-4s+3)}\right\} = \frac{4}{9}L^{-1}\left\{\frac{1}{s}\right\} + \frac{1}{3}L^{-1}\left\{\frac{1}{s^2}\right\} + \frac{1}{18}L^{-1}\left\{\frac{1}{s-3}\right\} - \frac{1}{2}L^{-1}\left\{\frac{1}{s-1}\right\}$$

Using the results  $L\{t^n\} = \frac{n!}{s^{n+1}}, n \geq 0$   $L\{e^{at}\} = \frac{1}{s-a}$

$$L^{-1}\left\{\frac{1}{s^2(s^2-4s+3)}\right\} = \frac{4}{9} + \frac{1}{3}t + \frac{1}{18}e^{3t} - \frac{1}{2}e^t \quad (2)$$

$$\frac{1}{s(s^2-4s+3)} = \frac{1}{s(s-3)(s-1)} = \frac{A}{s} + \frac{B}{s-3} + \frac{C}{s-1}$$

As  $s$ ,  $(s-3)$  and  $(s-1)$  are linear factors, we have

$$A = \frac{1}{s^2-4s+3}\bigg|_{s=0} = \frac{1}{3} \qquad B = \frac{1}{s(s-1)}\bigg|_{s=3} = \frac{1}{3(3-1)} = \frac{1}{6}$$

$$C = \frac{1}{s(s-3)}\bigg|_{s=1} = \frac{1}{1(1-3)} = -\frac{1}{2}$$

$$\begin{aligned}\frac{1}{s(s^2 - 4s + 3)} &= \frac{1}{3} \frac{1}{s} + \frac{1}{6} \frac{1}{s-3} - \frac{1}{2} \frac{1}{s-1} \\ L^{-1} \left\{ \frac{1}{s(s^2 - 4s + 3)} \right\} &= \frac{1}{3} L^{-1} \left\{ \frac{1}{s} \right\} + \frac{1}{6} L^{-1} \left\{ \frac{1}{s-3} \right\} - \frac{1}{2} L^{-1} \left\{ \frac{1}{s-1} \right\} \\ &= \frac{1}{3} + \frac{1}{6} e^{3t} - \frac{1}{2} e^t\end{aligned}\quad (3)$$

Using (2) and (3) in (1), we have

$$\begin{aligned}y &= 6 \left[ \frac{4}{9} + \frac{1}{3}t + \frac{1}{18}e^{3t} - \frac{1}{2}e^t \right] - 8 \left[ \frac{1}{3} + \frac{1}{6}e^{3t} - \frac{1}{2}e^t \right] \\ &= \left[ \frac{8}{3} + 2t + \frac{1}{3}e^{3t} - 3e^t \right] - \left[ \frac{8}{3} + \frac{4}{3}e^{3t} - 4e^t \right] \\ &= 2t + \left( \frac{1}{3} - \frac{4}{3} \right) e^{3t} - (3-4)e^t = 2t - e^{3t} + e^t\end{aligned}$$

**Example:** Solve the following problem using Laplace Transform

$$y'' + y = 2t, \quad y\left(\frac{\pi}{4}\right) = \frac{\pi}{2}, \quad y'\left(\frac{\pi}{4}\right) = 2 - \sqrt{2}$$

**Solution:** In this problem the initial conditions refer to some later instant  $t = t_0$ .

Such problems are called **Shifted Data Problems**.

As the values  $y$  and its derivatives occurring in Laplace transform are not readily available, we may proceed to solve such problems in two ways:

First method: Taking Laplace transform of DE we get

$$s^2 L\{y\} - sy(0) - y'(0) + L\{y\} = \frac{2}{s^2}$$

$$(s^2 + 1)L\{y\} = sy(0) + y'(0) + \frac{2}{s^2}$$

$$L\{y\} = \frac{sy(0)}{s^2 + 1} + \frac{y'(0)}{s^2 + 1} + \frac{2}{s^2(s^2 + 1)}$$

$$\begin{aligned} \therefore y(t) &= y(0)L^{-1}\left\{\frac{s}{s^2 + 1}\right\} + y'(0)L^{-1}\left\{\frac{1}{s^2 + 1}\right\} + L^{-1}\left\{\frac{2}{s^2(s^2 + 1)}\right\} \\ &= y(0)L^{-1}\left\{\frac{s}{s^2 + 1}\right\} + y'(0)L^{-1}\left\{\frac{1}{s^2 + 1}\right\} + L^{-1}\left\{2\left(\frac{1}{s^2} - \frac{1}{s^2 + 1}\right)\right\} \end{aligned}$$

$$\therefore y(t) = y(0)\cos t + y'(0)\sin t + 2t - 2\sin t$$

$$y'(t) = -y(0)\sin t + y'(0)\cos t + 2 - 2\cos t$$

Applying the conditions  $y\left(\frac{\pi}{4}\right) = \frac{\pi}{2}, \quad y'\left(\frac{\pi}{4}\right) = 2 - \sqrt{2}$

$$\frac{\pi}{2} = y(0)\frac{1}{\sqrt{2}} + y'(0)\frac{1}{\sqrt{2}} + 2\frac{\pi}{4} - 2\frac{1}{\sqrt{2}}$$

$$2 - \sqrt{2} = -y(0)\frac{1}{\sqrt{2}} + y'(0)\frac{1}{\sqrt{2}} + 2 - 2\frac{1}{\sqrt{2}}$$

Solving these equations we get  $y(0) = y'(0) = 1$

$$\therefore y(t) = 2t + \cos t - \sin t$$

**Second Method:** Put  $t = t_1 + \frac{\pi}{4}$  then for  $t = \frac{\pi}{4}$ ,  $t_1 = 0$

$$y(t) = y(t_1 + \frac{\pi}{4}) = y_1(t_1)$$

$$y' = \frac{dy(t)}{dt} = \frac{dy_1(t_1)}{dt} = \frac{dy_1(t_1)}{dt_1} \frac{dt_1}{dt} = \frac{dy_1(t_1)}{dt_1} = y_1' \quad y'' = y_1''$$

The IVP takes the form

$$y_1'' + y_1 = 2\left(t + \frac{\pi}{4}\right), \quad y_1(0) = \frac{\pi}{2}, \quad y_1'(0) = 2 - \sqrt{2}$$

Taking the Laplace transform we get

$$s^2 L\{y_1\} - sy_1(0) - y_1'(0) + L\{y_1\} = \frac{2}{s^2} - \frac{\pi}{2s}$$

**Example:** Solve the following simultaneous differential equations  
using Laplace Transform

$$\frac{dx}{dt} + y = e^t, \quad \frac{dy}{dt} + x = \sin t$$

given that  $x = 1, y = 0$  when  $t = 0$ .

**Solution:** Taking the Laplace Transform of both equations

$$sL\{x\} - x(0) + L\{y\} = \frac{1}{s-1}$$

$$L\{x\} + sL\{y\} - y(0) = \frac{1}{s^2 + 1}$$

$$\text{or } sL\{x\} - 1 + L\{y\} = \frac{1}{s-1}$$

$$L\{x\} + sL\{y\} = \frac{1}{s^2 + 1}$$

$$\text{or } sL\{x\} + L\{y\} = \frac{1}{s-1} + 1$$

$$L\{x\} + sL\{y\} = \frac{1}{s^2 + 1}$$

Solving these equations  $L\{x\}$  and  $L\{y\}$ , we get

$$\text{or } sL\{x\} + L\{y\} = \frac{1}{s-1} + 1$$

$$L\{x\} + sL\{y\} = \frac{1}{s^2 + 1}$$

$$L\{x\} = \frac{1}{s^2 - 1} \left( \frac{s}{s-1} + s - \frac{1}{s^2 + 1} \right)$$

$$L\{y\} = \frac{1}{s^2 - 1} \left( \frac{s}{s^2 + 1} - \frac{1}{s-1} - 1 \right)$$

$$L\{x\} = \frac{s}{(s-1)(s^2 - 1)} + \frac{s}{s^2 - 1} - \frac{1}{(s^2 + 1)(s^2 - 1)}$$

$$L\{y\} = \frac{s}{(s^2 - 1)(s^2 + 1)} - \frac{1}{(s^2 - 1)(s-1)} - \frac{1}{s^2 - 1}$$

$$\begin{aligned}
L\{x\} &= \frac{s}{(s-1)(s^2-1)} + \frac{s}{s^2-1} - \frac{1}{(s^2+1)(s^2-1)} \\
&= \frac{s}{(s-1)^2(s+1)} + \frac{s}{(s-1)(s+1)} - \frac{1}{(s^2+1)(s-1)(s+1)} \\
&= \frac{1}{2(s-1)^2} + \frac{1}{4(s-1)} - \frac{1}{4(s+1)} \\
&\quad + \frac{1}{2(s-1)} + \frac{1}{2(s+1)} - \frac{1}{4(s-1)} + \frac{1}{4(s+1)} + \frac{1}{2(s^2+1)} \\
&= \frac{1}{2(s-1)^2} + \frac{1}{2(s-1)} + \frac{1}{2(s+1)} + \frac{1}{2(s^2+1)} \\
x &= L^{-1} \left\{ \frac{1}{2(s-1)^2} + \frac{1}{2(s-1)} + \frac{1}{2(s+1)} + \frac{1}{2(s^2+1)} \right\} \\
x &= \frac{1}{2} \left\{ te^t + e^t + e^{-t} + \sin t \right\}
\end{aligned}$$

$$\begin{aligned}
L\{y\} &= \frac{s}{(s^2-1)(s^2+1)} - \frac{1}{(s^2-1)(s-1)} - \frac{1}{s^2-1} \\
&= \frac{s}{(s-1)(s+1)(s^2+1)} - \frac{1}{(s+1)(s-1)^2} - \frac{1}{(s-1)(s+1)} \\
&= \frac{1}{4(s-1)} + \frac{1}{4(s+1)} - \frac{s}{2(s^2+1)} \\
&\quad - \frac{1}{2(s-1)^2} + \frac{1}{4(s-1)} - \frac{1}{4(s+1)} - \frac{1}{2(s-1)} - \frac{1}{2(s+1)} \\
&= -\frac{s}{2(s^2+1)} - \frac{1}{2(s-1)^2} - \frac{1}{2(s+1)} \\
y &= L^{-1} \left\{ -\frac{s}{2(s^2+1)} - \frac{1}{2(s-1)^2} - \frac{1}{2(s+1)} \right\} \\
y &= -\frac{1}{2} \left\{ \cos t + te^t + e^{-t} \right\}
\end{aligned}$$

**Example:** Solve the following initial value problem using Laplace Transform

$$y'' + 3y' + 2y = r(t), \quad y(0) = 0, \quad y'(0) = 0$$

$$(a) \quad r(t) = u(t-1) - u(t-2) \quad \text{Square wave}$$

$$(b) \quad r(t) = \delta(t-1) \quad \text{Unit impulse}$$

**Solution:** Laplace transform of the DE is

$$s^2 L\{y\} - sy(0) - y'(0) + 3[sL\{y\} - y(0)] + 2L\{y\} = L\{r(t)\}$$

$$(s^2 + 3s + 2)L\{y\} = L\{r(t)\}$$

$$L\{y\} = \frac{1}{s^2 + 3s + 2} L\{r(t)\} \quad (1)$$

$$(a) \quad r(t) = u(t-1) - u(t-2) \quad L\{r(t)\} = \frac{e^{-s}}{s} - \frac{e^{-2s}}{s}$$

$$L\{y\} = \frac{1}{s(s^2 + 3s + 2)} (e^{-s} - e^{-2s}) = F(s)(e^{-s} - e^{-2s})$$

$$F(s) = \frac{1}{s(s^2 + 3s + 2)} = \frac{1}{s(s+2)(s+1)} = \frac{1}{2s} - \frac{1}{s+1} + \frac{1}{2(s+2)}$$

$$f(t) = L^{-1}\{F(s)\} = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}$$

$$y = L^{-1}\{F(s)(e^{-s} - e^{-2s})\} = f(t-1) \cdot u(t-1) - f(t-2) \cdot u(t-2)$$

$$= \begin{cases} 0 & 0 < t < 1 \\ f(t-1) & 1 < t < 2 \\ f(t-1) - f(t-2) & t > 2 \end{cases}$$

$$y = \begin{cases} 0 & 0 < t < 1 \\ \frac{1}{2} - e^{-(t-1)} + \frac{1}{2}e^{-2(t-1)} & 1 < t < 2 \\ -e^{-(t-1)} + \frac{1}{2}e^{-2(t-1)} + e^{-(t-2)} - \frac{1}{2}e^{-2(t-2)} & t > 2 \end{cases}$$

$$(b) \quad r(t) = \delta(t-1)$$

$$L\{r(t)\} = e^{-s}$$

$$L\{y\} = \frac{1}{s^2 + 3s + 2} e^{-s} = F(s)e^{-s}$$

$$F(s) = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

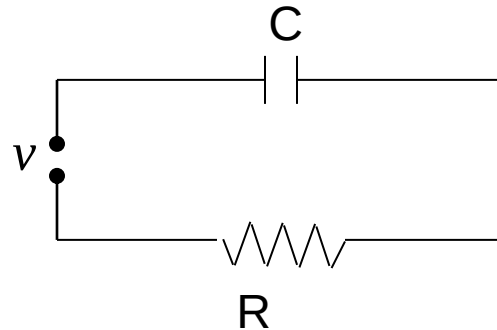
$$f(t) = e^{-t} - e^{-2t}$$

$$y = L^{-1}\{F(s)e^{-s}\} = f(t-1) \cdot u(t-1)$$

$$= \begin{cases} 0 & 0 \leq t < 1 \\ f(t-1) & t > 1 \end{cases}$$

$$= \begin{cases} 0 & 0 \leq t < 1 \\ e^{-t+1} - e^{-2(t-1)} & t > 1 \end{cases}$$

**Example:** Find the current  $i(t)$  in the circuit



if a single square wave with voltage  $V_0$  is applied. The circuit is supposed to be quiescent before square wave is applied.

**Solution:** The equation of the circuit is

$$v(t) = Ri(t) + \frac{1}{C} \int_0^t i(u) dt$$

$$v(t) = \begin{cases} V_0 & a \leq t \leq b \\ 0 & \text{otherwise} \end{cases} \quad v(t) = V_0[u(t-a) - u(t-b)]$$

Taking the Laplace transform

$$V(s) = RI(s) + \frac{I(s)}{Cs} \quad V(s) = V_0 \left[ \frac{e^{-as}}{s} - \frac{e^{-bs}}{s} \right]$$

$$I(s) = V_0 \left[ \frac{e^{-as}}{s} - \frac{e^{-bs}}{s} \right] \frac{1}{R + \frac{1}{Cs}} = V_0 \left[ \frac{e^{-as}}{s} - \frac{e^{-bs}}{s} \right] \frac{Cs}{RCs + 1}$$

$$I(s) = \frac{V_0}{R} [e^{-as} - e^{-bs}] \frac{1}{s + \frac{1}{RC}} = \frac{V_0}{R} [e^{-as} - e^{-bs}] F(s)$$

$$\begin{aligned} i(t) &= L^{-1}\{I(s)\} = \frac{V_0}{R} L^{-1}\{[e^{-as} - e^{-bs}]F(s)\} \\ &= \frac{V_0}{R} \left[ L^{-1}\{e^{-as}F(s)\} - L^{-1}\{e^{-bs}F(s)\} \right] \\ &= \frac{V_0}{R} [u(t-a)f(t-a) - u(t-b)f(t-b)] \end{aligned}$$

$$F(s) = \frac{1}{s + \frac{1}{RC}}$$

$$f(t) = e^{-\frac{t}{RC}}$$

$$\therefore i(t) = \frac{V_0}{R} \left[ u(t-a)e^{-\frac{t-a}{RC}} - u(t-b)e^{-\frac{t-b}{RC}} \right]$$

$$\therefore i(t) = 0 \quad \text{for} \quad 0 \leq t < a$$

$$i(t) = \frac{V_0}{R} e^{-\frac{t-a}{RC}} \quad \text{for} \quad a < t < b$$

$$i(t) = \frac{V_0}{R} \left[ e^{-\frac{t-a}{RC}} - e^{-\frac{t-b}{RC}} \right] = \frac{V_0}{R} \left[ e^{\frac{a}{RC}} - e^{\frac{b}{RC}} \right] e^{-\frac{t}{RC}} \quad \text{for} \quad t > b.$$

NEXT

**Thank you**